

Fault Detection and Data-driven Optimal Adaptive Fault-tolerant Control for Autonomous Driving using Learning-based SMPC

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Abstract—Real-time fault detection and active fault-tolerant control on vehicles can prevent potential accidents and ensure safe autonomous driving. Unlike existing methods which are designed for known faults or require perfect knowledge of vehicle models, this paper proposes a novel data-driven optimal adaptive fault-tolerant control method with model learning for automated vehicles to enable effective and safe driving under various faults. The actuator and sensor faults are integrated in one unified model to describe the vehicle dynamic behavior. The vehicle states are estimated online with only limited onboard sensors in vehicles available. An efficient fault detector using a modified sliding innovation sequence method is developed to ensure accurate fault detection. The dynamic model is adapted online using Gaussian Process based on vehicle running data, accurately representing the dynamic driving behavior. A novel learning-based adaptive fault-tolerant control method is designed to optimize the vehicle movement for autonomous path tracking under various faults with stochastic model predictive control. Simulation experiments validate the effectiveness of the proposed method in fault detection, model online learning, and accurate path tracking across various fault scenarios.

Index Terms—Adaptive fault-tolerant control, state estimation, fault detection, model online adaption, optimal path tracking.

I. INTRODUCTION

Autonomous driving technology has emerged as a prominent direction with the potential to enhance road traffic safety and efficiency [1]. The reliability and stability of autonomous driving systems have received considerable attention in both industry and academia. In automated vehicles (AVs), the growing number of sensors have increased the system complexity, making the vehicles susceptible to faults. The widely applied drive-by-wire technologies have also brought great demand in the system redundancy. When the vehicle suffers from faults on key components, it may lose stability and cause potential danger during driving without timely full control or effective takeover by human drivers. Fig.1 presents that in the event of actuator or sensor faults, AVs failed to drive on the desired path. It results in sudden deviation from the driving lane and lead to collisions

with surrounding vehicles. To solve these issues, the fault tolerance strategy can be applied on the system from the software level to ensure the normal operation when the vehicle face faults. This strategy aims to maintain the AVs controllability and stability under various fault issues [5][6]. Therefore, real-time fault detection and active fault-tolerant control (FTC) on vehicles play significant role to effectively prevent potential accidents and ensure safe autonomous driving.

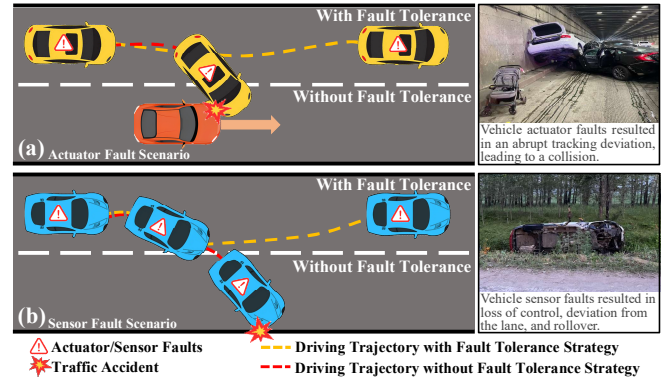


Fig.1. The influence of faults on safe autonomous driving.

FTC is getting intensive research attention for improved performance and stability, which can be divided into two categories, passive FTC and active FTC. Passive FTC utilizes a fixed fault-tolerant controller and predefined control laws for specific faults without considering real-time detection. It is therefore unable to work effectively in complex systems [7]. In contrast, active FTC integrates a fault detector and a reconfigurable controller [8-11]. Fault detection (FD) aims to estimate the fault severity and location. Based on the detection results, the active FTC method can adaptively reconfigure parameters to address faults [12].

Two dominant FD approaches, state-estimation-based and learning-based methods, have been extensively investigated. Learning-based methods, including neural networks [13][14], deep learning [15], and reinforcement learning [16], require substantial computational resources, posing challenges in real application. While state-estimation-based methods detect faults by comparing estimated values with sensor readings, which

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relies on accurate physical model. However, the accuracy and reliability of the model will affect the performance of fault detection [17-19].

Extensive research has been conducted on active FTC strategies. A novel event-triggered adaptive PID controller was developed for FTC of underactuated autonomous surface vehicles, with consideration for state saturation [20]. For unmanned aerial vehicles (UAVs), an active FTC strategy was proposed, combining nonlinear observer technology for fault diagnosis with sliding mode control (SMC) for fault tolerance [21]. A robust adaptive SMC method was employed to ensure that the system can recover to a stable state within a finite time under actuator faults or uncertainties [22][22]. In addition, a FTC method was proposed for four in-wheel motor driven electric vehicles, featuring torque redistribution in case of motor failure to achieve a balance between vehicle stability and power consumption [24]. A method for FTC of quadcopters' attitude under sensor faults and disturbances was devised, which eliminated the sensor faults estimated by sliding mode observers [25]. Notably, Most of the work above require prior knowledge of specific faults and accurate system dynamic models [26-28]. This is generally difficult to get during vehicle running. Furthermore, they only focus on a single type of faults with good performance in specific faults, but the methods are unable to work properly on unknown various fault scenarios. Thus, a more advanced FTC method is imperative to improve the reliability and generality of the control performance.

Machine learning (ML)-based methods have advantages in handling multiple faults without reliance on precise dynamic models and have been successfully applied in FTC. The reinforcement learning-based FTC method can address unknown faults in nonlinear systems [29][30]. ML-based methods have been applied in various fields including robotic control [31], battery energy management [32], and driver behavior [33]. However, the variability of faults may result in limited adaptability of learning-based methods and it is difficult to apply ML-based FTC algorithms for safe driving in diverse fault scenarios.

Conventional path tracking algorithms, such as model predictive control (MPC) and linear quadratic regulator, have limited performance in uncertain dynamic systems due to their reliance on fixed vehicle models. Particularly they are unable to address path tracking problems of AVs under unknown faults. Stochastic MPC (SMPC) uses probabilistic constraints for state and control variables to address system uncertainties [34][35]. SMPC ensures robust vehicle tracking ability under uncertainties, including navigation with obstacles and lane change risk management [36][37]. However, few studies have considered the impact of faults on optimal path tracking of AVs.

To summarize, one fundamental challenge in achieving accurate path tracking for AVs under unknown faults is dealing with the complex vehicle dynamic behavior in various fault scenarios. The current methods in the above literature have the following limitations:

- 1) The accuracy of vehicle state variables plays an important role in fault detection and tolerance control. However, the

accurate state estimation under faults with limited onboard sensors for autonomous driving has not been studied yet.

- 2) A large number of existing path tracking algorithms rely on the fixed vehicle model to derive the control policy, which is not applicable in uncertain conditions under various faults.
- 3) Current FTC methods for vehicles under faults only focus on solving a robust control problem, and they do not consider the optimal path tracking control for AVs to realize safe and stable driving under sensor and actuator faults.

To cover above issues, we propose a novel framework integrating fault detection and optimal learning-based adaptive FTC (LAFTC) for safe autonomous driving under various unknown faults, as shown in Fig. 2. The proposed framework enables real-time system state estimation and accurate fault detection using only limited onboard sensors available in vehicles. The system dynamics model can be updated online using running data to capture the varying dynamic behavior under faults, and is integrated in the optimal adaptive FTC for safe autonomous driving to realize accurate path tracking. The proposed method can effectively deal with optimal motion control of AVs when facing various faults.

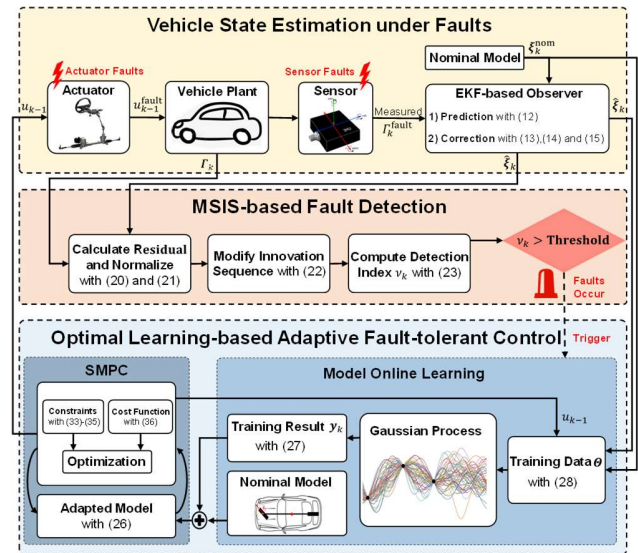


Fig. 2. The framework of the fault detection and optimal learning-based adaptive fault-tolerant control.

The main contributions are summarized as follows:

- 1) The vehicle state variables under actuator and sensor faults are estimated online with limited onboard sensors available in vehicles (e.g. IMU). The novel state estimation method can provide precise state information for fault detection and feedback control in the adaptive FTC strategy during autonomous path tracking.
- 2) An effective fault detector with the modified sliding innovation sequences (MSIS) method is proposed for accurate fault detection. It can efficiently detect both actuator and sensor faults and achieve the highest performance metric compared to other conventional fault detectors. When the faults are accurately detected, the proposed LAFTC strategy is triggered.
- 3) An online optimal LAFTC strategy for accurate and safe autonomous path tracking is developed with SMPC. The

vehicle model can be adapted online using running data to illustrate different vehicle dynamic behaviors under various faults, enabling to handle various fault cases for safe driving. It can improve the tracking performance with less lateral error compared to two other robust control methods.

To the best knowledge of the authors, this is the first data-driven optimal adaptive FTC method with model online learning for autonomous driving to ensure driving safety and reliability. The codes are open-sourced under <https://github.com/mobility-23/Fault-Tolerant-Control-System>.

The rest of the paper is organized as follows. Section II presents the vehicle dynamic model and faults model, followed by an introduction to state estimation based on extended Kalman filter (EKF). Section III presents the real-time fault detection method with MSIS in detail. Section IV introduces the LAFTC algorithm for path tracking of AVs. In Section V, extensive simulation experiments are conducted to illustrate the effectiveness of the proposed FTC strategy. Conclusions are drawn in Section VI.

II. VEHICLE DYNAMIC MODELING AND STATE ESTIMATION

In this section, the unified vehicle dynamic model under actuator and sensor faults is derived. First a nonlinear system model to describe the vehicle movement and the path tracking model for motion control are presented. Vehicle common faults on actuators and sensors studied in this paper are introduced. The state estimation of the vehicle dynamic system is presented with EKF for fault detection and adaptive FTC.

A. Vehicle Dynamic Model

A monorail vehicle model is utilized to describe the vehicle movement. Fig. 3 shows the schematic of the vehicle system. The vehicle model can be described by:

$$\ddot{x} = \dot{\phi} \dot{y} + a_x \quad (1a)$$

$$\ddot{y} = -\frac{2C_{ar} + 2C_{af}}{mV_x} \dot{y} + \left(-V_x - \frac{2C_{af}l_f - 2C_{ar}l_r}{mV_x}\right) \dot{\phi} + \frac{C_{af}}{m} \delta_f \quad (1b)$$

$$\ddot{\phi} = -\frac{2C_{af}l_f - 2C_{ar}l_r}{I_z V_x} \dot{y} + \frac{2C_{af}l_f^2 + 2C_{ar}l_r^2}{I_z V_x} \dot{\phi} + \frac{2C_{af}l_f}{I_z} \delta_f \quad (1c)$$

where δ_f is the front steering angle, x is the longitude position of the vehicle, y is the lateral position of the vehicle and ϕ is the yaw angle, a_x and V_x represent the longitudinal acceleration and longitudinal velocity, m is the vehicle mass, C_{af} and C_{ar} are the stiffness of the front and rear tires, I_z denotes the yaw moment of inertia, l_f and l_r are the distance from the center of gravity (CG) of the vehicle to the front and rear axles.

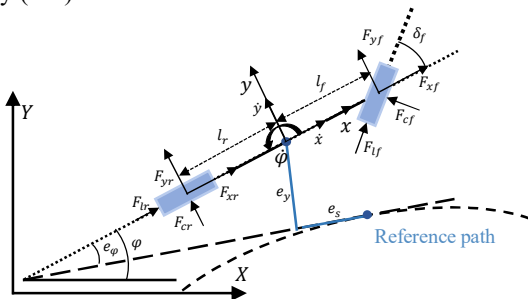


Fig. 3. Schematic of the vehicle model.

When the AV follows a desired path, an error model for path tracking is shown as:

$$\dot{e}_y = \dot{y} + V_x e_\psi \quad (2a)$$

$$\ddot{e}_y = \ddot{y} + V_x \dot{e}_\psi \quad (2b)$$

$$\dot{e}_\phi = \dot{\phi} - \dot{\phi}_{des} \quad (2c)$$

$$\ddot{e}_\phi = \ddot{\phi} - \ddot{\phi}_{des} \quad (2d)$$

$$\dot{e}_s = V - V_{des} \quad (2e)$$

$$\dot{e}_v = a - a_{des} \quad (2f)$$

where e_y represents the lateral error, e_ϕ denotes the heading error, while $\dot{e}_y$ and $\dot{e}_\phi$ represent their respective rates. Specifically, e_y signifies the distance between the vehicle's CG and the orthogonal projection point of the CG on the desired path, e_s and e_v denotes the longitudinal distance error and longitudinal velocity error with respect to reference values, ϕ_{des} , V_{des} and a_{des} are the reference yaw angle, reference velocity and reference acceleration, respectively.

The output of the system is designed as:

$$\dot{y} = \dot{e}_y - V_x e_\phi \quad (3a)$$

$$\ddot{y} = \ddot{e}_y - V_x \dot{e}_\phi \quad (3b)$$

$$\dot{\phi} = \dot{e}_\phi + \dot{\phi}_{des} \quad (3c)$$

Combining (1) - (3), the discretized system dynamic model for path tracking in the state-space representation can be described by:

$$\begin{cases} \xi_k = f(\xi_{k-1}, \mathbf{u}_{k-1}) + \omega_{k-1} \\ \mathbf{r}_k = h(\xi_k) + \mathbf{v}_k \end{cases} \quad (4)$$

where the state of the AV system is denoted as $\xi = [e_y, \dot{e}_y, e_\phi, \dot{e}_\phi, e_s, e_v]^T \in \mathcal{R}^{n_\xi \times 1}$, $\mathbf{u} = [\delta_f, T]^T$ is the control input and $T = a - a_{des}$, while δ_f and $T \in \mathcal{R}^{n_u \times 1}$, $\mathbf{r} = [\dot{y}, \ddot{y}, \dot{\phi}]^T$ and $\dot{y}, \ddot{y}, \dot{\phi}$ are lateral velocity, lateral acceleration and yaw rate, respectively, $\omega_{k-1} \sim \mathcal{N}(0, \mathbf{Q})$ and $\mathbf{v}_k \sim \mathcal{N}(0, \mathbf{R})$ are the process noise and measurement noise with gaussian distribution, f and h are the state equation and output equation, respectively.

B. Vehicle Common Faults during Operation

AVs may face different faults during operation. Accurate modeling of faults is critical to deal with associated risks, such as loss of control and inadequate regulation of vehicle movement. In general, the primary faults are categorized into two groups: actuator faults and sensor faults [38].

a. Modeling for Actuator Faults

Vehicle actuators are susceptible to faults caused by wear and tear, mechanical failures, electrical issues, or control system errors. When such faults happen, they may affect the performance of the actuators, resulting in a discrepancy between the actual control input $\mathbf{u}^{\text{fault}}$ applied on the vehicle and the designed control policy $\mathbf{u}$. Thus, the vehicle dynamics under the actuator fault can be described as:

$$\mathbf{u}_{k-1}^{\text{fault}} = \lambda \mathbf{u}_{k-1} + \boldsymbol{\tau} \quad (5a)$$

$$\boldsymbol{\xi}_k^{\text{fault}} = f(\boldsymbol{\xi}_{k-1}, \mathbf{u}_{k-1}^{\text{fault}}) \quad (5b)$$

where $\boldsymbol{\tau}$ denotes a fixed bias fault, such as being stuck at fixed output or an additive output, λ is the gain coefficient, $\boldsymbol{\xi}_k^{\text{fault}}$ corresponds to the vehicle state variables under actuator or sensor faults.

There are two main forms of actuator faults:

Fault 1: Constant gain faults: $\boldsymbol{\tau} = 0$

$$\mathbf{u}^{\text{fault}} = \lambda \mathbf{u} \quad (6)$$

It is characterized by the actuator whose actual control output $\mathbf{u}^{\text{fault}}$ is proportional to the desired control effort $\mathbf{u}$. Steering faults have a more substantial impact on both driving safety and vehicle operation, potentially leading to graver consequences. In this study, only the steering faults are considered here which is significant for the autonomous path tracking control.

Fault 2: Constant output faults: $\lambda = 0$

$$\mathbf{u}^{\text{fault}} = \boldsymbol{\tau} \quad (7)$$

Under this fault, the actuator maintains an unknown constant control effort, indicating that the actuator is nearly uncontrollable. This may cause the driving failure completely in the dynamic environment and it has no meaning for the FTC from the software level. Therefore, this fault is not considered in this paper.

b. Modeling for Sensor Faults

The sensors on vehicles are vulnerable to faults due to sensor degradation, calibration errors, signal noise, electromagnetic interference or environmental conditions affecting sensor performance, ultimately resulting in inaccurate sensor readings. Here it is assumed that the sensor fault is larger than the measurement noise. Consequently, in the presence of sensor fault, the system output will be given by:

$$\mathbf{r}^{\text{fault}} = h(\boldsymbol{\xi}_k) + \boldsymbol{\beta} \quad (8)$$

where $\mathbf{r}^{\text{fault}}$ represents the measurement of the vehicle system under faults, $\boldsymbol{\beta}$ is defined as a constant deviation in the sensor output from the true value. Lateral velocity is commonly one of the key factors that affect vehicle control, and it has direct implications for vehicle stability during path tracking. Therefore, we assume that a sensor fault causes a deviation in the lateral velocity measurement with $\boldsymbol{\beta} = [\eta, 0, 0]^T$. Here η is the bias value of the fault. Note that the proposed method in this paper can also be applied to solve sensor faults on other variables.

In summary, an integrated system state-space equation to describe vehicle dynamic behaviors under both actuator and sensor faults is derived through combining (3)-(8), which is formulated by:

$$\begin{cases} \boldsymbol{\xi}_k^{\text{fault}} = f(\boldsymbol{\xi}_{k-1}, \lambda \mathbf{u}_{k-1}) + \boldsymbol{\omega}_{k-1} \\ \mathbf{r}_k^{\text{fault}} = h(\boldsymbol{\xi}_k^{\text{fault}}) + \mathbf{v}_k + \boldsymbol{\beta} \end{cases} \quad (9)$$

When considering only actuator faults, $\boldsymbol{\beta} = [0, 0, 0]^T$, $\lambda \neq 1$; when considering only sensor faults, $\boldsymbol{\beta} = [\eta, 0, 0]^T$, $\eta \neq 0$, $\lambda = 1$.

C. Vehicle State Estimation with EKF

The nonlinear state-space equation in (4) to describe the actual vehicle driving behavior on normal conditions without faults is presented as:

$$\begin{cases} \boldsymbol{\xi}_k^{\text{nom}} = f(\boldsymbol{\xi}_{k-1}, \mathbf{u}_{k-1}) + \boldsymbol{\omega}_{k-1} \\ \mathbf{r}_k = h(\boldsymbol{\xi}_k^{\text{nom}}) + \mathbf{v}_k \end{cases} \quad (10)$$

where $\boldsymbol{\xi}_k^{\text{nom}}$ denotes the state variables of the nominal model. In this paper, it is assumed that the output vector $\mathbf{r} = [\dot{y}, \ddot{y}, \dot{\phi}]^T$ are measurable by IMU, which is available in AVs. The nonlinear dynamic model in (10) can then be linearized using Taylor series at the current operating point, which is shown as:

$$\boldsymbol{\xi}_k^{\text{nom}} = \mathbf{A}_d \boldsymbol{\xi}_{k-1} + \mathbf{B}_d \mathbf{u}_{k-1} + \mathbf{C}_d \dot{\phi}_{\text{des}} + \boldsymbol{\omega}_{k-1} \quad (11)$$

$$\mathbf{A}_d = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & \mathbf{A}_{d1} & \mathbf{A}_{d2} & \mathbf{A}_{d3} & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & \mathbf{A}_{d4} & \mathbf{A}_{d5} & \mathbf{A}_{d6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{B}_d = \begin{bmatrix} 0 & 0 \\ \frac{2C_{\alpha f}}{m} & 0 \\ 0 & 0 \\ \frac{2C_{\alpha f} l_f}{I_z} & 0 \\ 0 & 0 \\ 0 & -1 \end{bmatrix} \quad \mathbf{C}_d = \begin{bmatrix} 0 \\ -\frac{2C_{\alpha f} l_f - 2C_{\alpha r} l_r}{mV_x} - V_x \\ 0 \\ -\frac{2C_{\alpha f} l_f^2 + 2C_{\alpha r} l_r^2}{I_z V_x} \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{A}_{d1} = -\frac{2C_{\alpha f} + 2C_{\alpha r}}{mV_x}, \mathbf{A}_{d2} = \frac{2C_{\alpha f} + 2C_{\alpha r}}{m}, \mathbf{A}_{d3} = -\frac{2C_{\alpha f} l_f + 2C_{\alpha r} l_r}{mV_x}$$

$$\mathbf{A}_{d4} = -\frac{2C_{\alpha f} l_f - 2C_{\alpha r} l_r}{I_z V_x}, \mathbf{A}_{d5} = \frac{2C_{\alpha f} l_f - 2C_{\alpha r} l_r}{I_z}, \mathbf{A}_{d6} = -\frac{2C_{\alpha f} l_f^2 + 2C_{\alpha r} l_r^2}{I_z V_x}$$

where $\mathbf{A}_d \in \mathcal{R}^{n_\xi \times n_\xi}$, $\mathbf{B}_d \in \mathcal{R}^{n_\xi \times n_u}$ and $\mathbf{C}_d \in \mathcal{R}^{n_\xi \times 1}$, $\mathbf{H}_d \in \mathcal{R}^{n_\xi \times n_\xi}$ is a skew-diagonal matrix, ϕ_{des} denotes the reference path.

The EKF method is applied here to integrate system dynamics with sensor measurements to obtain optimal estimations of unknown system state variables. The process of state estimation is outlined as follows:

1) *Initial value selection*: The initial state value $\hat{\boldsymbol{\xi}}_0$ and the initial covariance matrix $\mathbf{P}_0$ are set during system initialization as the starting point for state estimation.

2) *Prediction*: Utilizing the estimated state and the control input at the preceding time step $k-1$, the current state and covariance matrix are given as:

$$\hat{\boldsymbol{\xi}}_{k|k-1} = \mathbf{A}_d \hat{\boldsymbol{\xi}}_{k-1} + \mathbf{B}_d \mathbf{u}_{k-1} + \mathbf{C}_d \dot{\phi}_{\text{des}} + \boldsymbol{\omega}_{k-1} \quad (12a)$$

$$\mathbf{P}_{k|k-1} = \mathbf{F}_{k-1} \mathbf{P}_{k-1} \mathbf{F}_{k-1}^T + \mathbf{Q} \quad (12b)$$

where $\hat{\boldsymbol{\xi}}_{k|k-1}$ is the state prior estimate at time k , $\mathbf{P}_{k|k-1}$ is the covariance prior estimate at time k , $\mathbf{F}_{k-1}$ is the state transition matrix, $\mathbf{Q}$ is the covariance matrix of process noise.

3) *Correction*: The process involves computing the Kalman gain coefficient for the current time step:

$$\mathbf{Z}_k = \mathbf{H}_d \mathbf{P}_{k|k-1}^- \mathbf{H}_d^T + \mathbf{R} \quad (13a)$$

$$\mathbf{k}_k = \mathbf{P}_{k|k-1}^- \mathbf{H}_d^T \mathbf{Z}_k^{-1} \quad (13b)$$

where $\mathbf{k}_k$ is the Kalman gain matrix at time step k . $\mathbf{R}$ is the covariance matrix of measurement noise.

Then, the estimated covariance matrix and the state variables based on the measurement Ψ_k from IMU are updated by:

$$\hat{\xi}_k = \hat{\xi}_k^- + \mathbf{k}_k(\Psi_k - \mathbf{H}_d \hat{\xi}_k^-) \quad (14a)$$

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{k}_k \mathbf{H}_d) \mathbf{P}_{k|k-1}^- \quad (14b)$$

where $\hat{\xi}_k$ and $\mathbf{P}_k$ are the posterior state estimate and covariance estimate at time step k . $\mathbf{F}_{k-1}$ and $\mathbf{H}_d$ are computed by:

$$\mathbf{F}_{k-1} = \frac{\partial \mathbf{F}(\xi)}{\partial \xi} = \mathbf{A}_d \quad (15a)$$

$$\mathbf{H}_d = \frac{\partial h(\xi)}{\partial \xi} = \begin{bmatrix} 0 & 1 & -V_X & 0 & 0 & 0 \\ 0 & 0 & 0 & -V_X & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \quad (15b)$$

Combining (12)-(15), the estimated system state with EKF can be obtained by:

$$\begin{aligned} \hat{\xi}_k &= \mathbf{A}_d \hat{\xi}_{k-1} + \mathbf{B}_d \mathbf{u}_{k-1} + \mathbf{C}_d \varphi_{\text{des}} + \omega_{k-1} \\ &+ \mathbf{k}_k(\Psi_k - \mathbf{H}_d(\mathbf{A}_d \hat{\xi}_{k-1} + \mathbf{B}_d \mathbf{u}_{k-1} + \mathbf{C}_d \varphi_{\text{des}} + \omega_{k-1})) \end{aligned} \quad (16)$$

III. FAULT DETECTION DESIGN

In this section, A MSIS-based detector is applied to efficiently diagnose whether the vehicle is under fault. The input of the detector consists of system state vector ξ_k , control input $\mathbf{u}_k$ for tracking and measurements Ψ_k from sensors. The detection index is calculated through the differences between the actual measurements and the expected outputs over a period of time. If the detection index exceeds the specified threshold, it indicates that faults have occurred in the vehicle.

The MSIS detector encompasses two stages: the calibration phase and the detection phase [39].

1) Calibration Phase:

Due to dynamic variations in AVs during complex maneuvers, the estimated states may have large estimation errors at the beginning of the maneuver. To obtain the accurate estimated states, The state estimation can be optimized through the predetermined sliding window, with a duration specified by the time window parameter:

$$\hat{\xi}_k = \underset{\xi_t, \mathbf{u}_t}{\operatorname{argmin}} \sum_{t=\max(1, k-n+1)}^k (\hat{\xi}_t - \xi_t) \quad (17)$$

where n is the length of the sliding window. The calibration phase enables the detector to attain precise estimations of the state and the covariance matrix to obtain the optimal $\hat{\xi}_k$. Converting the the Eq. (13b) and Eq. (14a), the process of the calibration phase can be expressed as:

$$\mathbf{k}_k^i = \mathbf{P}_{k|k-1}^- \mathbf{H}_d^T (\mathbf{H}_d \mathbf{P}_{k|k-1}^- \mathbf{H}_d^T + \mathbf{R})^{-1} \quad (18a)$$

$$\hat{\xi}_k^{i+1} = \hat{\xi}_k^i + \mathbf{k}_k^i (\Psi_k - \mathbf{H}_d \hat{\xi}_k^i) \quad (18b)$$

where $i \in \{0, 1, \dots, M-1\}$ represents the iteration step. The $\hat{\xi}_k^i = \hat{\xi}_k$ when $i = 0$. $\mathbf{k}_k^i$ and $\hat{\xi}_k^i$ are the Kalman gain matrix and state variables in the iteration step. The iteration procedure stops when the discrepancy between $\hat{\xi}_k^{i+1}$ and $\hat{\xi}_k^i$ is below the predefined threshold σ . Upon completion of the iteration process, the current $\hat{\xi}_k^i$ is regarded as the optimal estimated state value $\hat{\xi}_k$.

2) *Detection Phase*: By applying a general operator norm to normalize the innovation sequence within a sliding window, the modified sliding innovation sequences are obtained.

Unlike conventional detectors, the MSIS detector continuously updates the innovation sequence within a fixed sliding window. The detector constructs a normalized random variable following the innovation sequence, conforming to a probability distribution. Detection results are obtained by comparing the index to a predefined threshold.

The innovation sequence, representing the gap between the measured value and the expected value, can be formulated as:

$$\mathbf{z}_k = \Psi_k - h(\hat{\xi}_k) \quad (19)$$

where $\mathbf{z}_k \sim \mathcal{N}(0, \Phi_{\mathbf{z}_k})$. $\Phi_{\mathbf{z}_k} = (\mathbf{H}_d \mathbf{P}_{k|k-1}^- \mathbf{H}_d^T + \mathbf{R})$.

According to (4) and (16) - (19), $\mathbf{z}_k$ can be derived as:

$$\begin{aligned} \mathbf{z}_k &= h(\xi_k) + \mathbf{v}_k - h(\hat{\xi}_k) \\ &= h(\mathbf{A}_d \xi_{k-1} + \mathbf{B}_d \mathbf{u}_{k-1} + \mathbf{C}_d \varphi_{\text{des}} + \omega_{k-1}) + \mathbf{v}_k \\ &\quad - h(\mathbf{A}_d \hat{\xi}_{k-1} + \mathbf{B}_d \mathbf{u}_{k-1} + \mathbf{C}_d \varphi_{\text{des}} + \omega_{k-1}) \\ &\quad + \mathbf{k}_k(\Psi_k - \mathbf{H}_d(\mathbf{A}_d \hat{\xi}_{k-1} + \mathbf{B}_d \mathbf{u}_{k-1} + \mathbf{C}_d \varphi_{\text{des}} + \omega_{k-1})) \end{aligned} \quad (20)$$

The expression for the normalized innovation of ζ_k can be formulated as:

$$\zeta_k = \Phi_{\mathbf{z}_k}^{-\frac{1}{2}} \mathbf{z}_k' \quad (21)$$

The modified innovation sequence is denoted as ζ_k , and the set of modified sliding innovation sequences is defined as $S(\boldsymbol{\vartheta}_k) := \{\boldsymbol{\vartheta}_k \mid k \in [1, \infty]\}$. The expression for $\boldsymbol{\vartheta}_k$ is given as follows:

$$\boldsymbol{\vartheta}_k = \{\vartheta_{\max(1, k-n+1)}, \vartheta_{\max(1, k-n+1)+1}, \dots, \vartheta_k\} \quad (22)$$

where $\{\vartheta_t = q(\zeta_t) \mid t \in [\max(1, k-n+1), k]\}$. $q(\cdot)$ is used to calculate the operator norm to ζ_t .

The detection index ζ_k of the MSIS detector based on the sliding innovation sequence is expressed as:

$$\zeta_k = \frac{\gamma+1}{n} \sum \|\vartheta_t\|, \quad \vartheta_t \in \boldsymbol{\vartheta}_k \quad (23)$$

where γ is the relaxation factor, which is employed to calibrate the threshold of the detector. $\|\cdot\|$ represents 2-norms.

The decision boundary for the MSIS detector, characterized by a confidence level of $1 - \alpha$, can be defined as:

$$H_0: \zeta_k \leq \varrho, \quad \text{no faults} \quad (24a)$$

$$H_1: \zeta_k > \varrho, \quad \text{faults occur} \quad (24b)$$

where the detection threshold ϱ , which corresponds to the absolute value of the Z-score at the chosen confidence level,

can be represented as $\varrho = \left\| Z_{\frac{\varrho}{2}} \right\|$. The confidence level when choosing the detection threshold ϱ indicates the reliability of the selected value, expressed as a percentage (e.g., 95%). However, the threshold determined by theoretical principles tends to be conservative. In practical applications, the threshold often requires further refinement based on expert experience or repeated experiments as in [40].

Algorithm 1 outlines the detection process of the MSIS detector. Notably, both the time and space complexity of the MSIS detector are $O(n)$ because it only necessitates the storage and summation of the modified innovation sequence with a fixed-length sliding window of size n . Compared to other detectors (e.g. χ^2 and MEWMA detector) with a complexity of $O(k)$, the MSIS detector demonstrates significant advantages in terms of both time and space efficiency. This becomes especially noteworthy when we account for the increased innovation values that occur over time during real-time scenarios with faults. To achieve this, the MSIS detector employs sliding window technology, allowing it to capture recent abnormal signals and enhance overall detection accuracy. Selection of the sliding window length depends on data characteristics, detection objectives, and computational resources. It requires tuning through trial and errors. Note that the detection mechanism in Algorithm 1 is used to determine whether the fault happens. Unfortunately, it cannot distinguish what type of fault occurs.

Algorithm 1: The algorithm of modified sliding innovation sequences detection.

Input:

$\mathbf{u}_k, \hat{\xi}_k, \Psi_k$

Output:

Fault detection result

```

1: for  $k = 1$  to  $N$ 
2:   Calculate the residual value  $\mathbf{z}_k$  with (19)
3:   Construct a sliding innovation sequence  $\mathbf{s}_k$  with (21)
4:   Define  $\boldsymbol{\vartheta}_k$  with (22)
5:   Obtain the detection index  $\zeta_k$  with (23)
6:   if  $\zeta_k > \varrho$ , then
7:     Trigger alarm
8:     continue
9:   else
10:    continue
11: end for

```

IV. OPTIMAL LEARNING-BASED ADAPTIVE FAULT-TOLERANT CONTROL DESIGN

In this section, the novel optimal LAFTC strategy is designed to ensure the stability and safety of AVs during path tracking under actuator faults and sensor faults. The vehicle dynamic model under faults which is used for the motion control is adapted online based on GP with vehicle running data. When the faults are detected by the MSIS-based detector, the LAFTC method with SMPC is applied to derive the optimal control policy for the safe autonomous tracking. Constraints for safe and stable driving are analyzed.

A. Optimal Path Tracking Problem

In the path tracking, vehicles should move smoothly. The target is mainly to minimize deviations of the lateral error and the heading error with respect to the reference path. This ensures vehicles to follow the centerline of the road as closely as possible, while executing smooth trajectories. The optimal path tracking problem can be solved by the optimization method, which can be described in a general form with:

$$\min_{\xi_k, \mathbf{u}_{kk=0}} \sum_{k=0}^{\pi} c(\xi_k, \mathbf{u}_k) \quad (25a)$$

$$\text{s. t.} \quad \xi_k = \mathbf{A}_d \xi_{k-1} + \mathbf{B}_d \mathbf{u}_{k-1} + \mathbf{C}_d \dot{\varphi}_{\text{des}} + \boldsymbol{\omega}_{k-1} \quad (25b)$$

$$\xi_k, \mathbf{u}_k \in \mathcal{V} \quad (25c)$$

where π is the time duration, $c(\cdot)$ is the cost function, and $\mathcal{V}$ is the constraint imposed on the state and input.

B. Model Online Learning under Faults

When the vehicle suffers from faults, the real vehicle system dynamic behavior has a significant deviation compared to the one under the nominal condition. In this situation, the real state values of vehicle under faults are not known exactly, and the nominal model is not suitable for the accurate motion control anymore. To address this issue, the model online learning approach utilizing GP is implemented. This approach learns the differences between the nominal model and real state values of vehicle caused by faults. The model can then be updated for various cases to describe the current vehicle driving behaviors, which is applied to compute the optimal control policy for the FTC. Note that this method is effective for both actuator and sensor faults. The state values under sensor faults are estimated by (16). The real vehicle dynamic model under faults is then approximated and adapted by the combination of the nominal model part and the GP-based learned model part as:

$$\begin{aligned} \xi_k^{\text{ada}} &= \underbrace{\xi_k^{\text{nom}}}_{\text{nominal model part}} + \underbrace{\mathbf{B} \hat{\mathbf{y}}_k}_{\text{GP-based learned model part}} \\ &= \mathbf{A}_d \xi_{k-1} + \mathbf{B}_d \mathbf{u}_{k-1} + \mathbf{C}_d \dot{\varphi}_{\text{des}} + \boldsymbol{\omega}_{k-1} + \mathbf{B} \mathbf{y}_k \end{aligned} \quad (26)$$

where ξ_k^{ada} is the adapted state variables correspond to the adapted model. The matrix $\mathbf{B} \in \mathcal{R}^{n \times n_\xi}$ is a coefficient matrix that depends on the learned model $\mathbf{y}_k$. $\mathbf{y}_k$ can be described as

$$\mathbf{y}_k = d_{GP}(\boldsymbol{\varepsilon}_k) + \boldsymbol{\omega}_{GP} \quad (27)$$

where d_{GP} is an unknown function identified from training data which is modeled by GP. The entries $\boldsymbol{\varepsilon}_k = [\mathbf{B}_{\varepsilon_\xi} \xi_k, \mathbf{B}_{\varepsilon_u} \mathbf{u}_k]$ correspond to the selected components of the state variable ξ_k and the control input $\mathbf{u}_k$. The dimensionality of the learning problem can be reduced by introducing the matrices $\mathbf{B}_{\varepsilon_\xi} \in \mathcal{R}^{n_{\varepsilon_\xi} \times n}$, $\mathbf{B}_{\varepsilon_u} \in \mathcal{R}^{n_{\varepsilon_u} \times n}$. The training data consists of inputs $\mathbf{G}$ and outputs $\mathbf{Y}$, which are specifically shown as follows:

$$\boldsymbol{\theta} = \begin{cases} \mathbf{G} = [\xi_{k-1}^{\text{fault}}, \mathbf{u}_{k-1}] \\ \mathbf{Y} = \xi_k^{\text{fault}} - \xi_k^{\text{nom}} \end{cases} \quad (28)$$

A prior GP model is proposed based on the inputs included in the matrix $\mathbf{g}$, which can be defined by:

$$f_{GP}(\mathbf{g}) \sim \mathcal{GP}(m(\mathbf{g}), p(\mathbf{g}, \mathbf{g}')) \quad (29a)$$

where f_{GP} is the GP model, its mean and covariance matrix are given as:

$$m(\mathbf{g}) = E(f_{GP}(\mathbf{g})) \quad (29b)$$

$$p[(\mathbf{g}, \mathbf{g}') = E[(f_{GP}(\mathbf{g}) - m(\mathbf{g}))(f_{GP}(\mathbf{g}') - m(\mathbf{g}'))] \quad (29c)$$

At a given data point $(\mathbf{G}_k, \mathbf{Y}_k)$, the posterior distribution of the unknown function is a gaussian distribution with mean $\mu^d(\mathbf{g})$ and variance $\Sigma^d(\mathbf{g})$:

$$\mu^d(\mathbf{g}) = m(\mathbf{g}) + k(\mathbf{g}, \mathbf{G})(k(\mathbf{G}, \mathbf{G}) + \sigma^2 \mathbf{I})^{-1}(\mathbf{Y} - m(\mathbf{G})) \quad (30a)$$

$$\Sigma^d(\mathbf{g}) = k(\mathbf{g}, \mathbf{g}) - k(\mathbf{g}, \mathbf{G})(k(\mathbf{G}, \mathbf{G}) + \sigma^2 \mathbf{I})^{-1}k(\mathbf{G}, \mathbf{g}) \quad (30b)$$

The zero-mean function $m(\mathbf{g}) = 0$ and a squared exponential kernel (SE) are employed to determine the covariance function $k(\mathbf{g}, \mathbf{g}')$. The resulting expression for the covariance function is as follows:

$$k(\mathbf{g}, \mathbf{g}') = \sigma_f^2 \exp(-0.5(\mathbf{g} - \mathbf{g}')^T M^{-1}(\mathbf{g} - \mathbf{g}')) \quad (31)$$

where $\sigma_f^2 \in R^{1 \times n_\epsilon}$ represents the output variance. $M \in R^{n_\epsilon \times n_\epsilon}$ is a diagonal matrix, and $M = \text{diag}([l_1, l_2, \dots, l_n])$ denotes the length scale covariance matrix.

The training data Θ are applied to train the GP model through (28)-(31) to obtain $\mathbf{y}_k$, enabling the vehicle dynamic model to be updated online using (26)-(27) to adapt to the changes.

C. Optimal Adaptive Fault-tolerant Control for Path Tracking

The objective of the optimal adaptive FTC for AVs is to ensure the driving stability and safety in the path tracking process under various faults. The vehicle should follow the reference line on roads and keep driving. The cost function can be defined as:

$$c(\xi_k^{\text{ada}}, \mathbf{u}_k) = \|\xi_k^{\text{ada}} - \xi_k^{\text{ref}}\|_L^2 + \|\mathbf{u}_k\|_I^2 \quad (32)$$

where ξ_k^{ref} is the reference state, L and I are assigned weights that correspond to the states and inputs, respectively.

In path tracking, the vehicle should strive to follow the reference path, with the distance between the vehicle and the reference path being constrained. Meanwhile, the vehicle's longitudinal and lateral motion should also be considered. Furthermore, the range of control inputs is restricted due to limitations in the steering and acceleration. Additionally, to avoid aggressive driving behavior, the rate of acceleration and steering should be limited. In summary, the constraints for states and inputs, and rate constraints for inputs are applied as:

$$\xi_{k_{\min}} \leq \xi_k \leq \xi_{k_{\max}}, k = 0, 1, \dots, N \quad (33a)$$

$$\mathbf{u}_{k_{\min}} \leq \mathbf{u}_k \leq \mathbf{u}_{k_{\max}}, k = 0, 1, \dots, N - 1 \quad (33b)$$

$$\Delta \mathbf{u}_{k_{\min}} \leq \Delta \mathbf{u}_k \leq \Delta \mathbf{u}_{k_{\max}}, k = 0, 1, \dots, N - 1 \quad (33c)$$

Since the state variables ξ contains the uncertainty resulting from the potential variation in fault severity under different temporal and environmental conditions, the inequality constraints are introduced to account for the possibility of faults in autonomous path tracking driving tasks. The inequality

constraints include information regarding the domains of state and control input. The state and input constraints $\Xi \in R^{n_\xi}$ and $U \in R^{n_u}$ are modeled using chance-constrained formulations by:

$$Pr(\xi \in \Xi) \geq 1 - \epsilon^\xi \quad (34a)$$

$$Pr(\mathbf{u} \in U) \geq 1 - \epsilon^u \quad (34b)$$

where ϵ^ξ and ϵ^u are the probability of violation.

The probability constraint in (34) can be transformed into a linear constraint, which can be reformulated as:

$$H^\xi \mu^\xi + r^\xi \sqrt{H^\xi \Sigma^\xi (H^\xi)^T} \leq b^\xi \quad (35a)$$

$$H^u \mu^u + r^u \sqrt{H^u \Sigma^u (H^u)^T} \leq b^u \quad (35b)$$

where symbol r corresponds to the quantile function $\Phi^{-1}(1 - \epsilon)$ of the standard normal distribution, while the kernel H and the mean μ represent parameters of the Gaussian distribution.

In summary, the data-driven optimal LAFTC for path tracking with model learning using SMPC in the receding horizon can be expressed as follows:

$$\min_U \sum_{k=0}^{N-1} \left(\|\mu_k^{\text{ada}} - \xi_k^{\text{ref}}\|_L^2 + \|\mathbf{u}_k\|_I^2 \right) + \|\mu_N^{\text{ada}} - \xi_N^{\text{ref}}\|_P^2 \quad (36a)$$

$$\text{s. t. } \xi_k^{\text{ada}} = \mathbf{A}_d \xi_{k-1} + \mathbf{B}_d \mathbf{u}_{k-1} + \mathbf{C}_d \dot{\varphi}_{\text{des}} + \omega_{k-1} + \mathbf{B} \mathbf{y}_k \quad (36b)$$

$$\xi_k \sim \mathcal{N}(\mu^{\xi_k}, \Sigma^{\xi_k}) \quad (36c)$$

$$\xi_{k_{\min}} \leq \xi_k \leq \xi_{k_{\max}} \quad (36d)$$

$$\mathbf{u}_{k_{\min}} \leq \mathbf{u}_k \leq \mathbf{u}_{k_{\max}} \quad (36e)$$

$$\Delta \mathbf{u}_{k_{\min}} \leq \Delta \mathbf{u}_k \leq \Delta \mathbf{u}_{k_{\max}} \quad (36f)$$

$$Pr(\xi_k \in \Xi) \geq 1 - \epsilon^\xi \quad (36g)$$

$$Pr(\mathbf{u}_k \in U) \geq 1 - \epsilon^u \quad (36h)$$

$$\mu_0^\xi = \xi_k \quad (36i)$$

Where μ_k^ξ is the mean of the state variable, N is the prediction horizon, P is the weighting matrix of the terminal states. The first element in the optimized control input sequence is applied on the vehicle to accurately track the path even under faults.

The algorithm of the LAFTC is outlined in Algorithm 2, which provides a step-by-step procedure for executing the key components of the LAFTC method.

Algorithm 2: Learning-based adaptive fault-tolerant control

- 1: Initialize time step k
 - 2: Initialize vehicle position $([x, y, \varphi])$
 - 3: **for** $k \leq \text{End time}$
 - 4: Update vehicle state ξ and control input $\mathbf{u}$
 - 5: Initialize the states associated to fault detection
 - 6: Perform state estimation based on EKF with (16)
 - 7: Detect faults with (24)
 - 8: **While** faults are detected from time step $k = n$ **do**
 - 9: Trigger alarm
 - 10: Collect $\xi_{k-1}^{\text{fault}}, \xi_k^{\text{fault}}, \xi_k^{\text{ref}}$ and $\mathbf{u}_{k-1}$ and store them
 - 11: **if** $k = n + i$ (i is a pre-defined time domain)
-

```

12:   Define the training dataset with (28)
13:   Train with GP and store the training results  $y_k$ 
      with (27)
14: end if
15: for  $n + i < k \leq$  End time
16:   Update the vehicle model with  $d_{GP}$  with (26)
17:   Switch the algorithm to the LAFTC method
18:   The LAFTC controls the vehicle to track the
      desired path during faults with (36)
19: end for
20:   Update the vehicle states
21: end while
22: end for

```

V. SIMULATION RESULTS AND DISCUSSION

In this section, various simulation tests are conducted to evaluate the performance of the state estimation, model online learning and fault detection. Furthermore, the path tracking scenarios under actuator and sensor faults are constructed to verify the optimal adaptive fault-tolerant performance for autonomous driving using the proposed LAFTC. The details of the faults are given in section II -B. Simulation tests are carried out in MATLAB, with the vehicle parameters listed in Table I. The system state-space formulation in (9) is used to describe the vehicle dynamic behaviors under sensor and actuator faults in the simulation. Here the model learning performance and the adaptive FTC for actuator and sensor faults are discussed separately to evaluate its robustness and adaptivity in various conditions. A circular track is employed as the driving scenario. The optimal path tracking with the nominal MPC method is used as baseline for comparison. Single-lap tracking scenario is used to evaluate the autonomous tracking performance based on the proposed LAFTC method, while multiple-laps tracking illustrates the model online adaption and optimal adaptive FTC when facing various faults.

TABLE I
VEHICLE PARAMETERS

Parameters	Symbol	value
Vehicle mass	m	2050 kg
Vehicle yaw inertia	I_z	3344 kg*m ²
stiffness of the front tires	C_{af}	80000 N/rad
stiffness of the rear tires	C_{ar}	80000 N/rad
Distance from CG to the front axle	l_f	1.43 m
Distance from CG to the rear axle	l_r	1.47 m

The driving scenario for autonomous tracking is illustrated in Fig. 4 which comprises three straights and seven corners of varying curvatures. The initial state of the vehicle is set as $[0,2,0]$. The solid black lines depict the road boundary, while the dashed black line represents the center line of the road. The purple line indicates the simulation results of the autonomous

tracking for the vehicle motion control without faults using MPC. As depicted in this figure, in the absence of faults, the optimized vehicle tracking trajectory in the purple line successfully follows the center line, which demonstrates the efficiency of the conventional MPC method for the optimal vehicle motion control in the absence of faults.

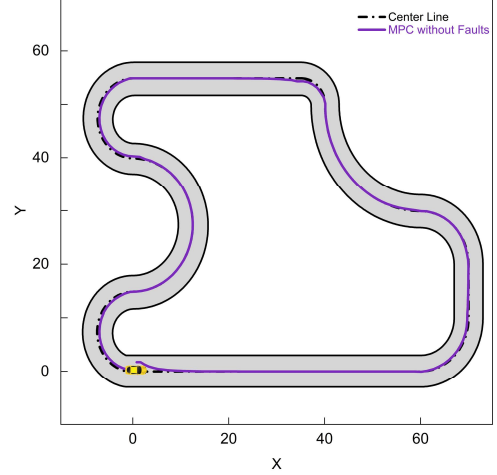
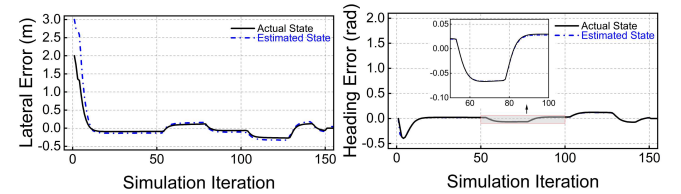


Fig. 4. Optimized vehicle trajectory during path tracking.

A. Performance on Vehicle State Estimation

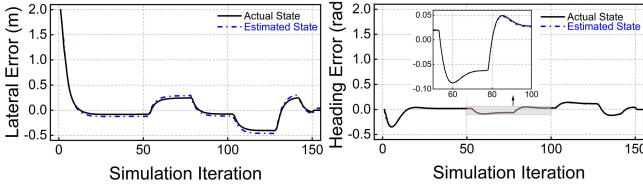
Assuming an initial deviation with $[0,2,0]$, the state estimation method proposed in this paper can effectively estimate the system's state variables using limited onboard sensors. The performance of the proposed EKF-based observer for the state estimation are demonstrated in Fig. 5 and Fig. 6. In both figures, the actual state variables in the black line represent the real vehicle states, while the estimated state pertains to the ones obtained by the observer shown in the blue. The initial value of the estimated state in the observer is set to $[0,3,0]$. The results in Fig.5 show that with the different initial position, the estimated value of the lateral error can quickly converge to the actual lateral error of the vehicle, and the difference between two values are close to zero finally. For the heading error, the estimated value is nearly the same as the actual one in both straight and curve road. It indicates that the proposed estimation method can efficiently capture the dynamics of the vehicle with limited sensors and provide accurate estimation of real values.



(a) The estimated lateral error (b) The estimated heading error
Fig. 5. The estimation results with different initial points.

To further validate the effectiveness of the state estimation method under faults, the actuator fault with $\lambda = 0.6$ is analyzed. Fig. 6 illustrates the state estimation performance of the observer in terms of the lateral error and heading error. It can be seen that in the presence of the actuator fault, the estimated value of the lateral error in the blue line closely aligns with the real lateral error of the vehicle in the black. The difference of

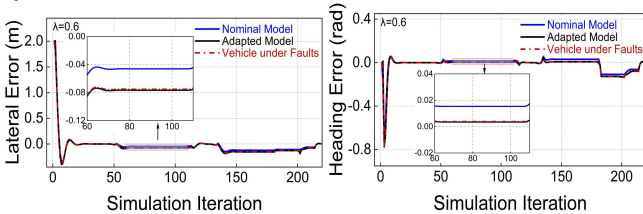
the heading error between the estimated value and the real state value is close to zero. It indicates that the full-state estimation is realized and the estimated state of the observer provide robust convergence to the actual state values of the vehicle under faults. Therefore, the above result emphasizes that the proposed EKF-based observer can effectively capture the system state values online. The estimated state can be used for the fault detection and adaptive FTC which are shown in the following parts.



(a) The estimated lateral error (b) The estimated heading error
Fig. 6. The estimation results with the actuator fault ($\lambda = 0.6$).

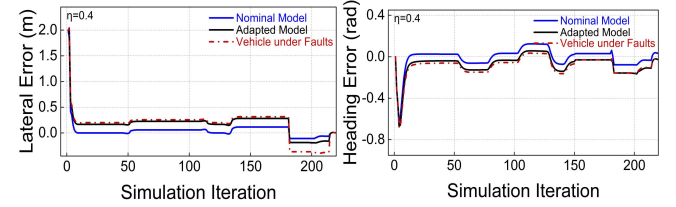
B. Performance on Model Online Learning under Faults

To evaluate the online learning ability of the adapted model, we compare the states derived from the nominal model, the adapted model, and the real states of the vehicle under different faults. The same control policy is applied here to propagate the system states for comparison of the three different models above. Here the nominal model serves as the baseline, which describes the vehicle dynamics without faults. The vehicle under faults represents the real vehicle states under actuator or sensor faults described using (9), while the adapted model is the model which is updated online with (26). The results of model online learning are illustrated in Fig. 7 and Fig. 8. The faults studied here contain the actuator fault with $\lambda = 0.6$ (in Fig. 7) and the sensor fault with $\eta = 0.4$ (in Fig. 8). As illustrated in Fig. 7, the actuator fault leads to both the considerable lateral error e_y and heading error e_ϕ in dash red line compared to the states of the nominal model without faults in the blue line. While the adapted model in the black line shows nearly the same lateral error and heading error with the states from the real vehicle under the actuator fault. This is because the nominal model fails to capture the dynamic changes caused by faults, leading to significant differences in model mismatch under faults. However, the adapted model can update the system dynamics resulting from the actuator fault with (26)-(27) using vehicle running data. It can therefore provide accurate model online learning performance and present the real vehicle dynamic behaviors.



(a) Later error comparison (b) Heading error comparison
Fig. 7. The model online learning performance of AV under the actuator fault with $\lambda = 0.6$.

To further validate the learning ability of the adapted model under the sensor fault, we compare the state values of three models as illustrated in Fig. 8. It is obvious that the sensor fault leads to noticeable discrepancies of the lateral error and the heading error between the nominal model and the actual vehicle state. It indicates that the nominal model is not applicable to present the actual operation of vehicle under sensor faults. However, the black solid line for the adapted model and the red dashed line from the actual vehicle state have almost the same trajectory. This is due to the model online learning using GP to effectively learn the dynamic changes caused by the sensor fault with vehicle running data, resulting in the state of the adapted model closely approximating the real state of the vehicle under faults. Therefore, the developed model online learning can effectively present the real vehicle dynamic behaviors under different running scenarios, which can be applied in the optimal adaptive FTC.



(a) Later error comparison (b) Heading error comparison
Fig. 8. The model online learning performance of AV under the sensor fault with $\eta = 0.4$.

C. Performance on Fault Detection with MSIS

The proposed MSIS detection method is evaluated on the vehicle under actuator faults with different severity ($\lambda = 0.6, 0.7, 0.8$, and 0.9). To demonstrate the effectiveness of the MSIS detector, it is compared with two commonly used fault detectors, i.e., χ^2 [39] and MEWMA [41]. Three metrics (*recall*, *precision* (*prec*) and *F1* score) are used to evaluated the detection performance in this paper. *Recall* calculates the ratio of true positives (TP) to all ground-truth positives, while *precision* calculates the ratio of TP over all identified positives [42]. Specifically, A higher *precision* value corresponds to a lower false positive (FP) rate, whereas a higher *recall* value corresponds to a higher TP rate. *F1* score is commonly employed to balance the detection performance of the detector in terms of FP and TP. The calculation of *F1* can be given as:

$$F1 = \frac{2 * precision * recall}{precision + recall} \quad (37)$$

Table II lists the results of different detection methods, with three key metrics. The evaluation of the detector's performance is primarily based on the *F1* score, which considers precision, recall, as well as the trade-off between false positives and false negatives. As is observed from Table II, the χ^2 detector exhibits high *precision* value, but suffers from the lowest *recall* value under multiple actuator faults, suggesting that the χ^2 detector can reliably identify faults but is prone to miss some faults. The MEWMA detector faces same challenges. In contrast, the proposed MSIS detector demonstrates the highest *recall* value,

implying minimal possibility of missed faults and improved fault detection capability. Furthermore, the MSIS detector achieves a high *prec* value, indicating low incidence of false positives. Overall, the MSIS method get the highest *F1* score, and the average *F1* score of MSIS detector under the four actuator faults is 0.039 and 0.027 higher than that of χ^2 detector and MEWMA detector, respectively. These findings consistently emphasize the effectiveness of the MSIS method, and it can accurately detect faults, issue alerts to trigger FTC module.

TABLE II
DETECTION PERFORMANCE COMPARISON OF
DIFFERENT DETECTORS

Actuator Fault	Method	<i>F1</i>	<i>Prec</i>	<i>Recall</i>
$\lambda = 0.6$	χ^2	0.9365	1	0.8806
	MEWMA	0.9624	0.9697	0.9552
	MSIS	0.9710	0.9434	1
$\lambda = 0.7$	χ^2	0.9280	1	0.8657
	MEWMA	0.9457	0.9839	0.9104
	MSIS	0.9635	0.9429	0.9451
$\lambda = 0.8$	χ^2	0.9163	0.9772	0.8626
	MEWMA	0.9193	1	0.8507
	MSIS	0.9545	0.9652	0.9438
$\lambda = 0.9$	χ^2	0.9016	1	0.8209
	MEWMA	0.9106	1	0.8358
	MSIS	0.9474	0.9525	0.9403

Fig. 9 gives further illustration on the fault detection by the MSIS detector during autonomous tracking. Assuming that the vehicle encounters actuator faults starting from step 54 of the simulation. The results demonstrate that for larger fault magnitudes, the fault can be promptly and accurately detected. However, decreasing fault magnitudes cause very minor detection delay due to indistinct initial sequences. Particularly, some abnormal ranges are observed in the fault detection window (red line), indicating that faults might be missed. This is because the steering command is zero when the vehicle drives on the straight road. For the straight driving, the vehicle is controlled by the tracking or braking force. Therefore, the faults on the steering actuator (discussed mainly in this paper) have no influence on the straight driving and could not be detected based on state values. When the vehicle drives on the curve road, the actuator fault can be observed based on the MSIS detector quickly and accurately. Otherwise, it will lead to a significant decline in detection performance. Overall, it can be concluded that the proposed MSIS detector can effectively detect the actuator faults in various fault conditions.

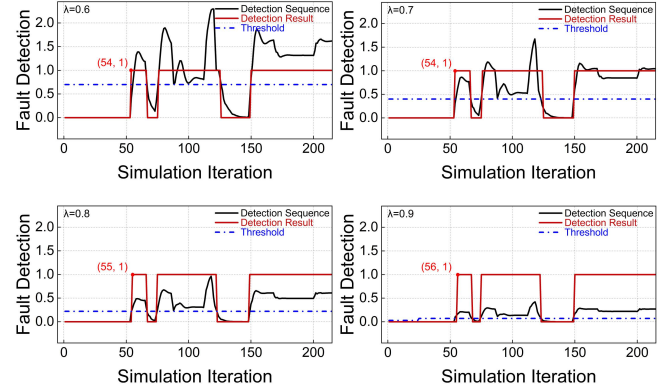


Fig. 9. Performance of the MSIS detector on actuator faults.

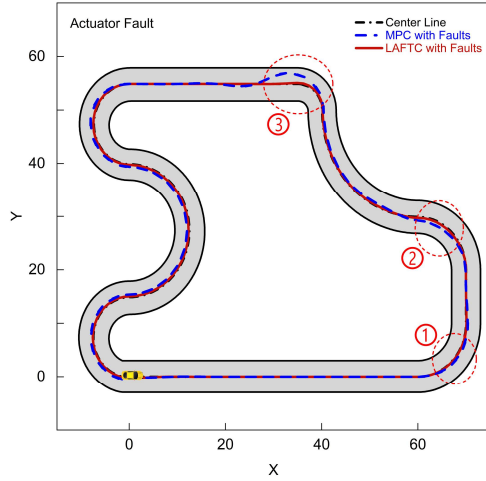
D. Optimal Adaptive Fault-tolerant Control for Path Tracking

The performance of the proposed optimal LAFTC method for path tracking is tested on the driving scenario in Fig. 4 to evaluate its capability for ensuring safe motion control under various fault scenarios. Path tracking performance can be evaluated by the lateral error and heading error, which reflect the deviation from the reference path and direction, respectively. The initial pose of the vehicle is set as $[0, 2, 0]$ and the vehicle motion under the conventional MPC method is used as the baseline. In this case, two scenarios, single-lap and multiple-laps tracking, are presented based on the running laps around the track. Tracking performance in the single-lap scenario is employed to evaluate the effect of the proposed fault tolerance strategy. Furthermore, the multiple-laps scenario is utilized to access the influence of model online learning with GP for the optimal FTC, highlighting the importance of the model adaption in various changing cases. Actuator and sensor faults have been considered in both tracking scenarios.

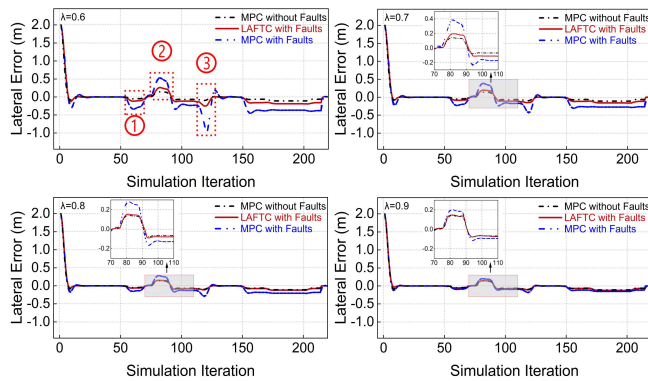
Single-lap tracking: In this scenario, AV completes a single lap under actuator faults with $\lambda = 0.6, 0.7, 0.8$ and 0.9 . The faults occur at the beginning of the path tracking process. Pre-trained GP is applied for FTC of the vehicle under different actuator faults. The scenario of sensor faults is the same with the ones for actuator faults in this test. Specifically, the fault parameter η is set to $-0.4, -0.2, 0.2$, and 0.4 , respectively. In this paper, the control effects of MPC method on the vehicle without faults and the vehicle under faults are denoted as MPC with faults and MPC without faults, respectively. Meanwhile, the control effect of the LAFTC method on the vehicle under faults is denoted as LAFTC with faults.

Fig. 10 displays the path tracking performance under actuator faults in the single-lap tracking. The overall tracking performance with the proposed adaptive FTC method is indicated in Fig. 10(a). With the conventional MPC method, actuator faults lead to a larger deviation of the vehicle's trajectory from the road center line particularly in regions ①-③ showing in the dashed blue line. It indicates that the conventional MPC method is unable to control the vehicle to follow the path effectively in the presence of actuator faults. The larger deviation at curves is caused by the characteristics of actuator faults mentioned in (5) and (6). Since the steering command on straightway is 0, this fault does not affect the

vehicle's motion on straightway but has a significant impact on the vehicle's motion at bends. However, the proposed LAFTC method can effectively reduce the trajectory deviation under fault conditions shown in the red line, and ensure accurate path tracking with the model learning. In Fig. 10(b), the comparison of the lateral error is presented with different actuator faults. The vehicle trajectory by the MPC method in the blue line is far from the reference line without faults in black. Noticeably, a larger fault magnitude (i.e. $\lambda = 0.6$) results in more lateral error, suggesting a bigger deviation of the vehicle from the center line of the road. This would cause possible driving danger in dynamic conditions. However, the value from the LAFTC method shown in the red line can closely track the reference line. This is due to the model online learning in (26), which learns the fault characteristics and can describe the real vehicle dynamics under faults. It enables the LAFTC control strategy to achieve better path tracking performance under the actuator fault conditions. The results demonstrate that the LAFTC method effectively improve the tracking of vehicle under actuator faults.



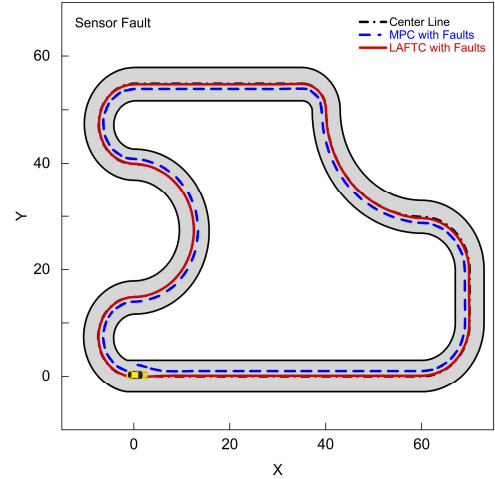
(a) Path tracking scenario under actuator fault.



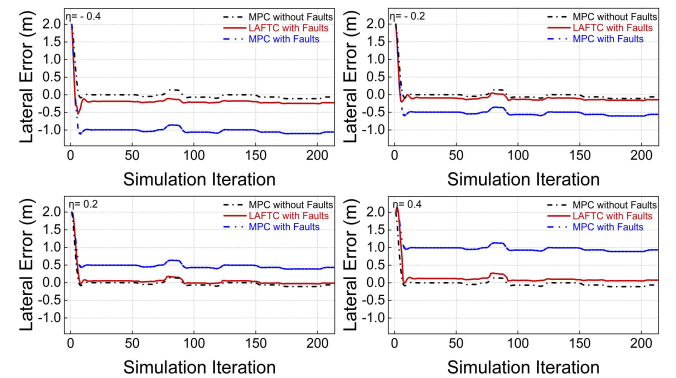
(b) Lateral error under different fault conditions.

Fig. 10. The path tracking performance under various actuator faults in single-lap tracking 1: (a) the scenario of path tracking under the actuator faults; (b) the comparison of the control algorithms on the lateral error under different actuator faults.

The vehicle tracking performance under sensor faults in the single-lap tracking is given in Fig. 11. As illustrated in Fig. 11(a), the trajectory under the MPC method, represented by the blue line, consistently deviates from the center line denoted in black. The conventional MPC method fails to control the vehicle with poor path tracking performance under the sensor fault. In contrast, the trajectory under the LAFTC method, depicted by the red line, closely adheres to the center line. It can provide accurate path tracking for safe autonomous driving. Further comparison of the lateral error with different sensor faults is provided in Fig. 11(b). The value by the LAFTC method in red closely track the reference line in black across diverse sensor fault scenarios. The conventional MPC method in blue results in much larger deviation from reference line, and larger fault magnitude leads to more lateral error. This illustrates the effectiveness and robustness of the proposed LAFTC method in accommodating and mitigating various sensor faults. The comprehensive analysis and comparison presented in these figures show the LAFTC method's capability to improve the tracking performance of vehicles in the presence of sensor faults.



(a) Path tracking scenario under sensor fault.



(b) Lateral error under different fault conditions.

Fig. 11. The path tracking performance under various sensor faults in single-lap tracking: (a) the scenario of path tracking under the sensor faults; (b) the comparison of the control algorithms on the lateral error under different sensor faults.

To further present the advantage of the LAFTC method, it is compared with two robust control methods commonly applied in fault-tolerant strategies, i.e. fuzzy control (FC) and sliding mode control (SMC). The comparison of the lateral error for path tracking with different methods under the actuator fault are shown in Fig. 12. The lateral error in the orange line by the FC method presents larger deviation, indicating very poor tracking. The FC method is unable to adapt to the changes in the driving scenario and requires the knowledge of the actuator fault to design the fuzzy logic. While the lateral error from the SMC approach in the blue line leads to less deviation and is close with the reference in the black, but it is still larger compared to the one from LAFTC. The SMC approach has the ability for the robust control for possible fault tolerance, however, it relies on the perfect system dynamic model under fault, which is generally difficult to obtain. Another issue of SMC comes from the oscillation and it has significant effect on the stability of the autonomous driving. Through comparison, the proposed LAFTC method provides the best tracking performance with minimum lateral error, which illustrates its considerable ability for the adaptive fault-tolerant control.

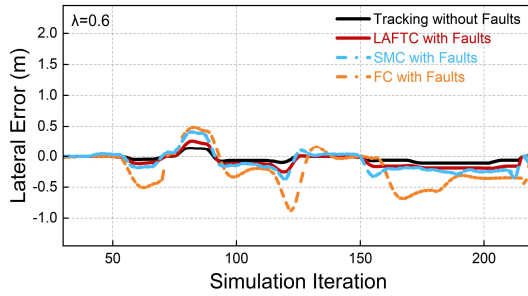


Fig. 12. The path tracking performance comparisons.

In order to further illustrate the optimal tracking performance of LAFTC, TABLE III presents the statistics comparisons of three methods in terms of the cumulative lateral error (CLE), average lateral error (ALE) and maximum lateral error (MLE). The lateral error value of LAFTC is used as the baseline. Compared with the LAFTC method, the SMC shows a significant increase in lateral errors, with its CLE increased by 83.6%, ALE increased by 90.0% and MLE increased by 142.3%, respectively. Furthermore, FC manifests the maximum lateral errors, leading to a 165.8% increase in CLE, a 170.0% increase in ALE, and a 238.5% increase in MLE, respectively, compared to LAFTC. This indicates that both SMC and FC lead to considerably more lateral error of the vehicle under faults, presenting poorer path tracking performance for autonomous driving. In conclusion, LAFTC outperforms traditional robust methods in optimal path tracking with fault tolerance, showcasing stronger robustness. It can improve the safe and reliable autonomous driving for vehicles running under faults.

TABLE III
COMPARISON OF DIFFERENT APPROACHES

Method	CLE(m)	ALE(m)	MLE(m)
LAFTC	19.92	0.10	0.26
SMC	36.57(↑ 83.6%)	0.19(↑ 90.0%)	0.63(↑ 142.3%)

FC	52.95(↑ 165.8%)	0.27(↑ 170.0%)	0.88(↑ 238.5%)
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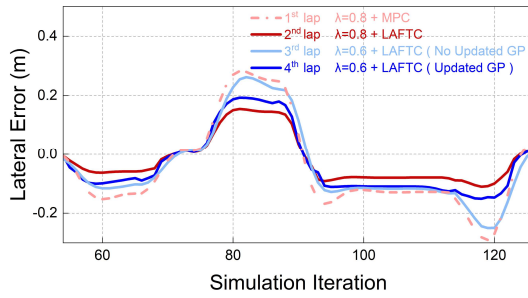
Multiple-laps tracking: AV completes multiple laps around the same sequence of the close track shown in Fig. 4 in the presence of actuator faults. In the 1st lap, the vehicle encounters the fault ($\lambda = 0.8$) and is controlled with the conventional MPC method. Model learning is conducted using GP after the 1st lap, and the fault is identified using the MSIS detector. Notably, the faults occur at step 54 of the 1st lap and has persisted since then. After detecting the fault, the LAFTC method is applied for the vehicle's motion control during the 2nd lap. In the 3rd lap, the vehicle experiences the actuator fault with $\lambda = 0.6$, and the LAFTC method uses the previous vehicle dynamic model trained in the 2nd lap without re-training the GP model or updating the vehicle dynamics model. In contrast, in the 4th lap, LAFTC is implemented with the new online updated system dynamic model. Once the GP model is retrained, the vehicle dynamics model is adapted again. Meanwhile, the sensor fault scenario is the same as the actuator fault scenario. It is notable that the vehicle tracks the prescribed trajectory with the sensor fault characterized by $\eta = 0.2$ during the 1st lap. Subsequently, during the 3rd lap, the AV encounters the sensor fault with $\eta = 0.4$. The faults occur at step 60 of the 1st lap. Table IV presents the faults and the models used in the controller for each lap.

TABLE IV
MODELS USED IN CONTROLLER FOR EACH LAP

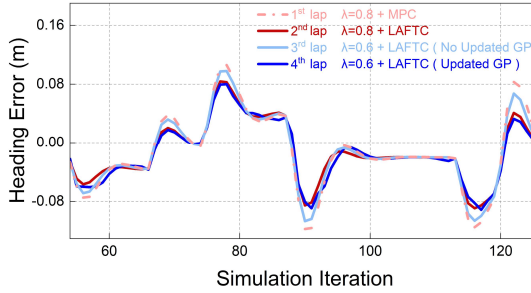
Fault Types	Laps	Fault Magnitude	Model
Actuator Faults	1 st lap	$\lambda = 0.8$	Nominal model
	2 nd lap		Adapted model (with GP-based learned model)
	3 rd lap	$\lambda = 0.6$	Same model used in 2 nd lap (No updated GP-based learned model)
	4 th lap		New adapted model (New online updated GP-based learned model)
Sensor Faults	1 st lap	$\eta = 0.2$	Nominal model
	2 nd lap		Adapted model (GP-based learned model)
	3 rd lap	$\eta = 0.4$	Same model used in 2 nd lap (No updated GP-based learned model)
	4 th lap		New adapted model (New online updated GP-based learned model)

Fig. 13 displays the comparison of the lateral error and the heading error for the path tracking performance of AV under actuator faults in multiple-laps. It can be seen that the dark red solid lines in both figures are noticeably smaller than the light red dashed lines, indicating that the lateral and heading errors in the 2nd lap have significantly decreased compared to the 1st lap. The proposed LAFTC method applied in the 2nd lap can mitigate the influence of faults and provides accurate tracking with model online learning with GP. When increasing the actuator fault in the 3rd lap, the old GP model trained from the 2nd lap is not suitable to present the dynamic driving behavior of the vehicle and the errors have been increased, resulting poor

tracking performance. However, in the 4th lap, the model is trained again with the new running data which can provide the current driving behavior in the new scenario. Compared with the 3rd lap in light blue line, the lateral error and heading error of the 4th lap in the dark blue line are significantly reduced. It illustrates that the LAFTC method with model online learning can give the excellent fault tolerance and improve the driving performance. It demonstrates strong adaptability in handling unexpected new actuator faults. Specifically, our algorithm utilizes an online adaptation model based on GP to enhance AV path tracking performance, ensuring safe operation. This indicates that our approach can effectively mitigate the impact of various actuator faults to optimize vehicle motion during path tracking.



(a) The lateral error of tracking under actuator faults.



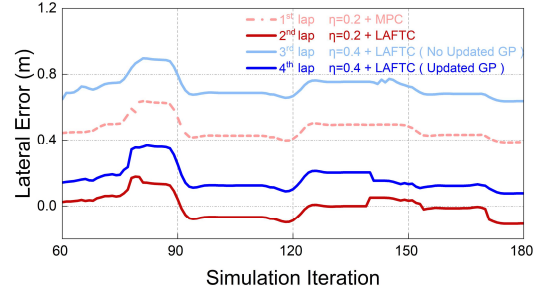
(b) The heading error of tracking under actuator faults.

Fig. 13. The path tracking performance of AV under actuator faults in multiple-laps tracking.

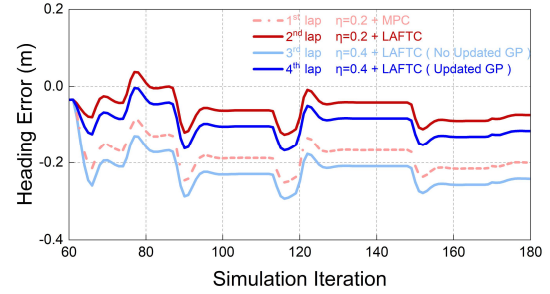
The path tracking performance in terms of the lateral error and the heading error of the AV under sensor faults during multiple-lap tracking is presented in Fig. 14. The sensor suffers from faults from step 60. The trajectory by the conventional MPC method showing in the light red dashed line leads to more errors in the 1st lap compared to the one in the 2nd lap where the LAFTC with model adaption is adopted for the motion control. It shows successful fault tolerance in the dark red solid line. When the fault changes in the 3rd lap, the old model is not applicable to describe the current vehicle dynamics, therefore the lateral error and the heading error increases. However, in the 4th lap, the vehicle model is adapted again with GP using the new running data, and it can provide accurate description for the vehicle dynamic behavior in the new fault. Therefore, the lateral error and the heading error in dark blue line has been reduced compared to the ones in the 3rd lap. Specifically, the proposed LAFTC algorithm leverages GP-based model online

adaptation to solve the optimal path tracking problem, which is adaptive to various sensor faults.

The above results highlight the adaptivity and robustness of the propose LAFTC method for excellent fault tolerance and improve the autonomous driving in changing and dynamic fault scenarios.



(a) The lateral error of tracking under sensor faults.



(b) The heading error of tracking under sensor faults.

Fig. 14. The path tracking performance of AV under sensor faults in multiple-laps tracking.

Table V presents the statistics of the cumulative and maximum lateral and heading errors observed during multiple laps under actuator and sensor faults. The findings reveal that in the 1st lap, there are higher cumulative and maximum values of lateral and heading errors. A significant error indicates poor path tracking performance, suggesting that the conventional MPC without model learning is unable to control the vehicle under faults. Conversely, the LAFTC method during the 2nd lap results in a substantial reduction of 86.0% in cumulative lateral errors and 84.1% in cumulative heading errors. Similarly, for sensor faults, 88.3% decrease in cumulative lateral error and 75.0% decrease in cumulative heading error are observed from the results. This demonstrates the LAFTC method's ability to substantially mitigate lateral and heading errors during vehicle path tracking, thereby enhancing overall tracking performance. Compared with the 3rd lap, the lateral and heading errors under actuator faults in the 4th lap is reduced by 41.6% and 50.4%, respectively. On the other hand, 82.4% and 60.1% reduction in cumulative lateral and heading errors are applied for the sensor faults, respectively. This indicates that the online adaptation of the system dynamic model can significantly reduce path tracking errors, enhancing the fault tolerance performance of the proposed LAFTC. Moreover, the changes in the maximum values follow the same trend as the cumulative values. These findings underscore the effectiveness of the LAFTC method in improving tracking performance in the presence of faults.

Additionally, the real-time updating of the dynamic model enhances the vehicle's ability to effectively manage variable faults. The analysis of vehicle tracking performance indicates that the LAFTC method can provide optimal tracking performance in various fault scenarios, demonstrating the potential of LAFTC to achieve safe and reliable autonomous driving in dynamic environments.

TABLE V
LATERAL AND HEADING ERROR UNDER FAULTS

Fault Types		Laps	Cumulative Value	Maximum Value
Actuator Faults	Lateral Error	1 st lap	5.461	0.193
		2 nd lap	0.766 (↓86.0%)	0.043
		3 rd lap	4.546	0.161
		4 th lap	2.656 (↓41.6%)	0.064
	Heading Error	1 st lap	0.860	0.049
		2 nd lap	0.137 (↓84.1%)	0.012
		3 rd lap	0.599	0.034
		4 th lap	0.297 (↓50.4%)	0.026
Sensor Faults	Lateral Error	1 st lap	25.708	0.496
		2 nd lap	3.016 (↓88.3%)	0.085
		3 rd lap	38.863	0.774
		4 th lap	6.819 (↓82.4%)	0.168
	Heading Error	1 st lap	8.821	0.179
		2 nd lap	2.201 (↓75.0%)	0.045
		3 rd lap	11.037	0.224
		4 th lap	4.404 (↓60.1%)	0.090

In this paper, it is assumed that the fault on the actuator and sensor only affects partial functionality, and the component does not fail to work completely. Therefore, the FTC strategy can be applied on the system from the software level to improve the driving safety when the vehicle suffers faults. If the fault causes the actuator or sensors failure totally, the redundant component from the hardware level should be designed in advance, which acts as a backup when the faulty component is isolated from the normal operation of vehicles. This will definitely increase the system complexity and cost, and it is not the focus of this paper.

VI. CONCLUSION

In this article, a novel learning-based optimal adaptive FTC method has been proposed to ensure safe and reliable driving for AVs under faults. First, the unified vehicle dynamic model considering actuator and sensor faults was derived to describe the vehicle motion under faults. Meanwhile, the vehicle state under faults was estimated online utilizing EKF. Furthermore, the MSIS detector was used to detect the actuator and sensor faults. When faults were detected, the model online learning mechanism was activated, incorporating GP to adapt to dynamic changes induced by various faults and update the vehicles' dynamics model in real-time. Subsequently, the data-driven optimal LAFTC method was employed to optimize the

autonomous path tracking of vehicles under faults. Finally, extensive simulations demonstrated the effectiveness of the proposed LAFTC method, with reduction of over 82% in cumulative lateral error and 60% in heading error with respect to the reference path across various fault scenarios. The proposed LAFTC method particularly enhances the robustness and safety of the vehicles' tracking performance in various fault scenarios. In the future, leveraging the advantage of the machine learning, the fault identification for different kinds of actuator and sensors will be explored to improve the safety and stability of autonomous driving. Furthermore, experimental tests will be conducted to evaluate the performance and robustness of the proposed LAFTC method for safe driving in terms of fault tolerance.

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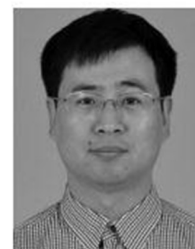


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